

SWAN ISLAND, Australia

WMO: 959850

Lat: 40.729S

Lon: 148.125E

Elev: 15

StdP: 101.14

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	6.5	-1.5	3.3	8.3	0.3	3.9	8.3	18.6	12.1	16.5	11.8	3.8	250	0.702

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.6	22.8	19.2	21.7	18.5	20.9	17.9	19.9	21.8	19.3	21.1	18.7	20.4	5.0	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.0	20.8	18.5	13.4	20.4	17.9	12.9	19.9	57.2	22.0	55.1	21.2	53.0	20.5	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	14.4	13.2	DB	3.7	26.4	0.7	2.0	3.2	27.8	2.8	29.0	2.5	30.1	2.0	31.5	
			WB	1.7	20.9	0.8	0.7	1.2	21.4	0.7	21.8	0.3	22.2	-0.3	22.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	17.6	17.9	17.2	15.3	13.1	11.4	10.6	10.5	11.4	12.7	14.3	16.0
	DBStd	3.17	1.90	1.77	2.09	1.92	1.87	1.72	1.49	1.51	1.50	1.67	1.84	1.81
	HDD10.0	35	0	0	0	0	1	6	11	12	4	1	0	0
	HDD18.3	1631	37	26	48	92	162	207	239	242	207	174	121	75
	CDD10.0	1491	235	222	222	158	98	49	30	29	47	85	130	186
	CDD18.3	43	13	14	11	1	0	0	0	0	0	0	1	3
	CDH23.3	28	13	6	6	0	0	0	0	0	0	0	1	2
	CDH26.7	2	2	1	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.7	6.9	6.6	6.4	6.3	6.7	6.4	6.9	6.8	6.9	6.8	6.8	7.1
Precipitation	PrecAvg	597	45	30	43	51	62	51	61	64	48	46	48	51
	PrecMax	1047	186	140	142	167	219	139	155	240	125	121	80	159
	PrecMin	379	6	0	2	2	9	20	9	15	15	7	6	2
	PrecStd	133	38	30	31	36	44	23	32	42	25	25	19	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.1	23.9	24.0	21.0	18.1	16.2	14.6	15.1	16.7	18.9	21.3	23.1
		MCWB	20.2	20.1	20.0	17.5	15.2	13.8	12.3	12.1	13.2	14.3	17.2	18.6
	2%	DB	22.7	22.5	22.1	19.8	17.2	15.2	13.9	14.1	15.3	17.1	19.3	20.8
		MCWB	19.1	19.2	18.8	17.0	14.7	13.1	11.7	11.7	12.6	13.7	16.3	17.4
	5%	DB	21.5	21.5	21.0	18.8	16.5	14.5	13.5	13.4	14.6	16.1	18.0	19.7
		MCWB	18.3	18.4	18.0	16.1	14.2	12.6	11.4	11.2	12.1	13.2	15.4	16.5
	10%	DB	20.5	20.6	20.1	18.1	15.9	14.0	13.0	12.9	13.9	15.3	17.1	18.8
		MCWB	17.5	17.8	17.6	15.6	13.7	12.1	11.0	10.8	11.5	12.6	14.5	15.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.2	20.7	20.6	18.6	16.3	14.8	12.9	12.7	14.0	15.5	18.2	19.2
		MCDB	22.8	22.6	22.9	19.7	17.0	15.5	14.0	13.7	15.2	17.2	20.0	21.8
	2%	WB	19.7	19.7	19.7	17.5	15.5	13.6	12.2	12.2	13.1	14.6	16.7	18.2
		MCDB	21.7	21.6	21.3	18.9	16.6	14.5	13.4	13.4	14.5	16.1	18.9	20.0
	5%	WB	19.0	19.2	19.0	16.8	14.7	12.9	11.8	11.6	12.5	13.9	15.8	17.4
		MCDB	21.0	20.9	20.7	18.4	16.1	14.1	13.0	13.0	13.9	15.6	17.6	19.1
	10%	WB	18.2	18.5	18.2	16.2	14.0	12.4	11.3	11.2	11.9	13.2	15.1	16.6
		MCDB	20.0	20.2	19.8	17.7	15.6	13.7	12.7	12.6	13.4	15.0	16.7	18.2
Mean Daily Temperature Range	5% DB	MDBR	4.9	4.6	4.7	4.5	4.2	3.8	3.9	4.3	4.5	4.7	4.7	4.8
		MCDBR	6.0	5.4	5.4	4.7	4.1	3.7	3.7	4.3	5.0	5.4	5.8	5.7
		MCWBR	3.8	3.5	3.5	3.3	3.2	2.8	2.8	3.1	3.4	3.7	3.7	3.7
	5% WB	MCDBR	5.6	5.0	4.9	4.4	3.6	3.3	3.3	3.8	4.3	5.0	5.4	5.4
		MCWBR	4.0	3.7	3.3	3.5	3.2	2.7	2.8	3.1	3.3	3.7	3.8	4.0
Clear-Sky Solar Irradiance	taub	0.349	0.345	0.335	0.321	0.308	0.302	0.300	0.312	0.333	0.335	0.359	0.344	
	taud	2.530	2.559	2.581	2.603	2.603	2.607	2.620	2.583	2.513	2.522	2.465	2.533	
	Ebn at Noon	979	958	925	872	817	789	819	869	911	957	963	990	
	Edh at Noon	109	100	90	76	65	59	63	76	94	103	115	109	
All-Sky Solar Radiation	RadAvg	6.82	5.85	4.63	3.17	2.13	1.68	1.90	2.77	4.05	5.38	6.31	7.08	
	RadStd	0.43	0.28	0.24	0.14	0.11	0.12	0.07	0.18	0.26	0.25	0.38	0.39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (15 neighbors)	+0.31	N/A	N/A	+0.72	+0.41	N/A	N/A	-78	+93	+14

Nomenclature: See separate page